



**Politecnico
di Torino**

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Survey on eMotors for electrical powertrains

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1 Introduction

1.1 Electric powertrain structures for the electric mobility

Electrified vehicles using an electric motor can be divided into seven categories:

- Micro Hybrid Vehicles
- Mild Hybrid Vehicles
- FHEV (Full Hybrid Electric Vehicles)
- PHEV (Plug-in Hybrid Electric Vehicles)
- EREV (Extended Range Electric Vehicles)
- BEV (Battery Electric Vehicles)
- FCEV (Fuel Cell Electric Vehicles)

In this list the categories are listed based on their dependence on the electric powertrain, starting from the one in which the E-motor has a marginal role, so the power is mostly provided by the ICE, to the one which uses only the electric motor for traction.

1.1.1 Micro Hybrid Vehicles

The Micro Hybrid Vehicles are those mounting only the so called "start/stop system", which helps to stop the combustion engine when the vehicle pulls to a stop and to restart it when the driver accelerates. Since the electric motor functionalities are limited to the start-stop, these type of vehicles mount an electric motor providing small value of power and a battery with a low value of capacity. The latter is charged by exploiting the regenerative braking or the kinetic energy of the vehicle during driving.

In Figure 1 the Audi Micro Hybrid architecture is represented. It is characterized by a BSG (Belt Starter Generator), i.e. the electric motor is connected to the ICE through a belt. This configuration is composed by an IPM motor supplied by a 12 V battery. The motor can provide a maximum power of about 7 kW and a torque of 50 Nm.

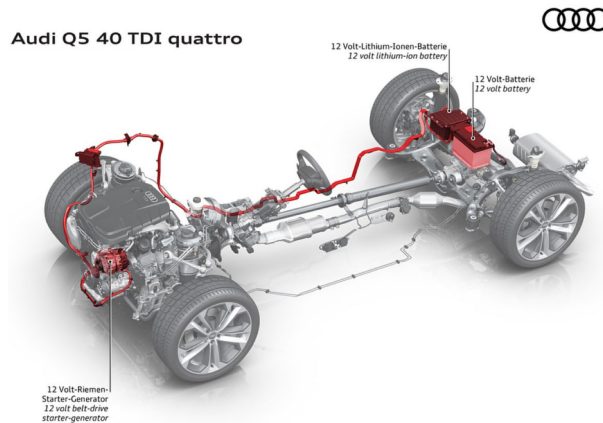


Figure 1: Audi Q5 Micro Hybrid architecture [12]

1.1.2 Mild Hybrid Vehicles

The Mild Hybrid Vehicles are equal to the Micro Hybrid ones, with the addition of the Electric Torque Assistance i.e. a supplement of torque, provided by the E-motor, when higher power is required, as during an overtaking. As for the Micro Hybrid Vehicles, the battery can be charged only during driving and with regenerative braking.

The Audi Mild Hybrid configuration, depicted in Figure 2, is characterized by a 15 kW and 100 Nm IPM motor, supplied by a 48 V battery.

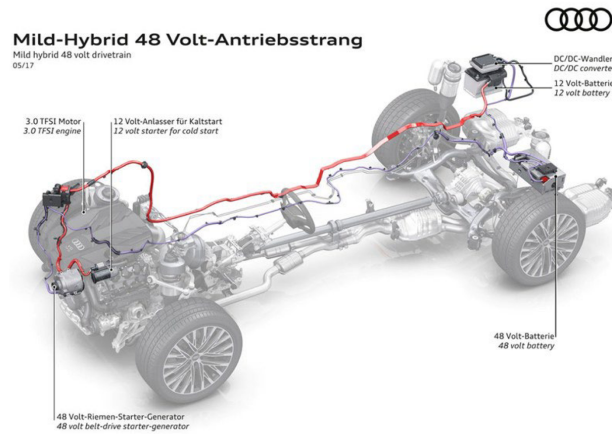


Figure 2: Audi Mild Hybrid powertrain [12]

1.1.3 Full Hybrid Vehicles

Full Hybrid Vehicles, unlike the previous ones, can run for short distances in pure electric, hence only by using the power provided by the electric motor. This means that they have to be equipped with a motor with a higher power values with respect to the one mounted on Micro and Mild Hybrid. Still the battery can be charged only through driving and regenerative braking. Obviously, these vehicles can still run only on the combustion engine, or even on a combination of both electric motor and thermal engine.

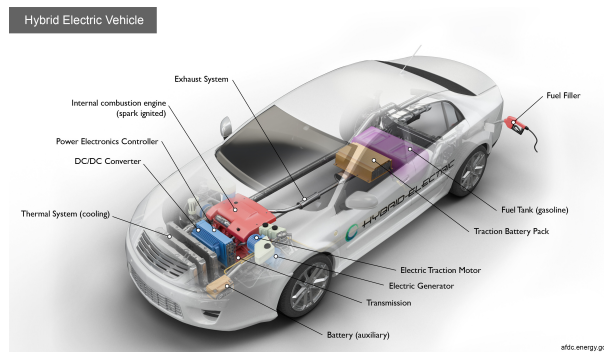


Figure 3: Full Hybrid architecture [7]

1.1.4 Plug-in Hybrid Vehicles

As Full Hybrid ones, even Plug-in Hybrid Vehicles can run only on the electric motor. The difference is that their battery can even be charged by plugging it into an external source of electric power, as well as during driving.

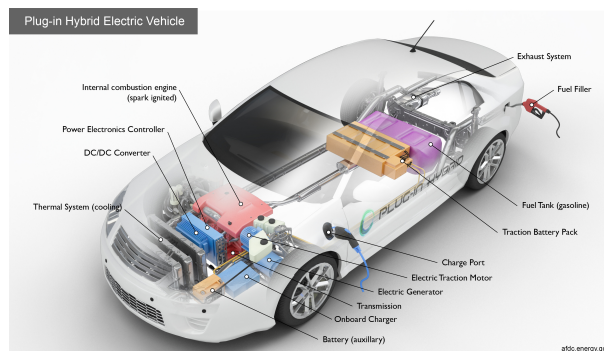


Figure 4: Plug-in Hybrid architecture [8]

1.1.5 Extended Range Electric Vehicles

The last category of vehicles with electric powertrain mounting also an ICE is the Extended Range Electric Vehicles category. The peculiar feature of these type of vehicles is that, even if equipped with an internal combustion engine, they run only on electric motor. Indeed, the ICE has the only purpose of charging the battery supplying the electric motor.

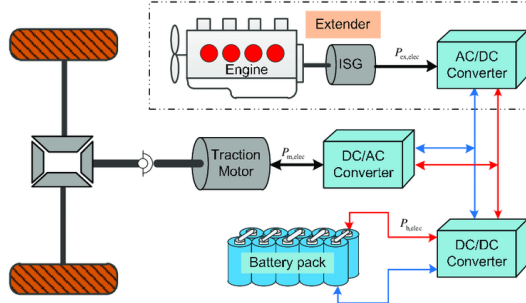


Figure 5: Extended Range Electric Vehicles architecture [1]

1.1.6 Battery and Fuel Cell Electric Vehicles

The last two categories include only vehicles fully powered by one, or more, electric motors. The difference between BEV and FCEV is that in Battery Electric Vehicles the electric energy needed to power the electric motor is supplied by the battery, while in Fuel Cell Electric Vehicles the energy is provided by the fuel cells, in particular by hydrogen.

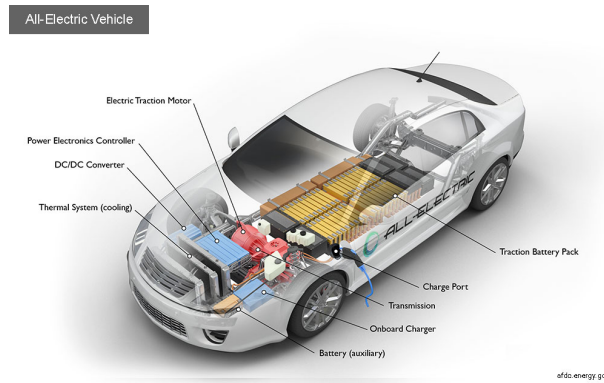


Figure 6: Full electric vehicle architecture [5]

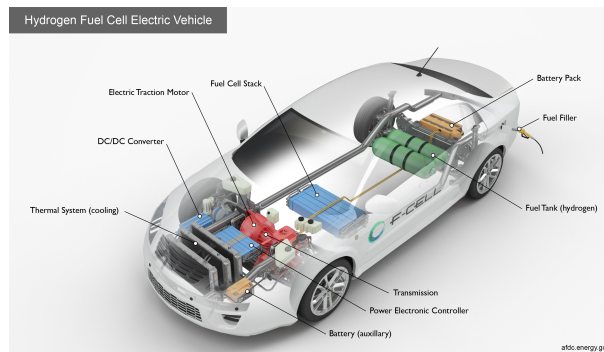


Figure 7: Fuel cell electric vehicle architecture [6]

An other important classification of the Hybrid Vehicles is based on the position, in the transmission, of the Electric Motor Drive (EMD), composed by the E-motor, the inverter and the control unit. The sorting is done considering eight categories, which are the following: P0, P1, P2, P2.5, PS, P3, P4 and P5, as shown in Figure 8.

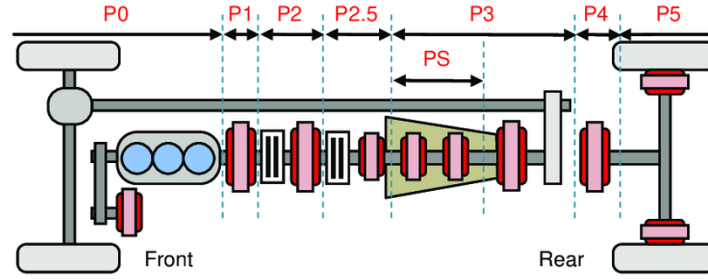


Figure 8: P-configurations for a HEV/EV [2]

1.1.7 P0

The P0 category is the one including the vehicles in which the EMD is side-attached with the ICE front through a belt. Usually this solution is used in the Micro and Mild Hybrid Vehicles. As a consequence, the electric motor and the battery are characterized by small values of, respectively, power and voltage.

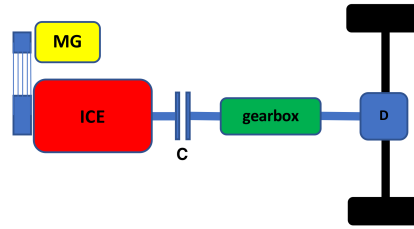


Figure 9: P0 configuration [13]

1.1.8 P1

In P1 vehicles, the EMD is directly connected on the ICE crankshaft, upstream of the clutch. This means that the rotation of the E-motor always depends on the internal combustion engine one, and viceversa. If one between ICE and EM is rotating, even the other one is rotating too. Thus, the vehicles mounting an electric motor in position P1 cannot run in full electric. The E-motor is only used to provide the Electric Torque Assistance, hence P1 is usually mounted on Mild Hybrid Vehicles.

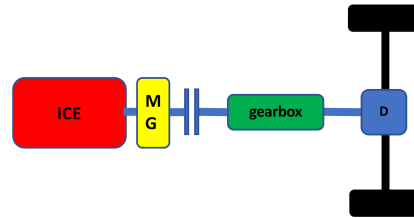


Figure 10: P1 configuration [13]

1.1.9 P2

The P2 category represents the vehicles mounting the EMD between ICE and transmission, downstream the clutch and upstream the gearbox. Since the electric motor can be either connected or disconnected from the ICE, the vehicles mounting a P2 configuration can run on full electric, as well as only on the combustion engine. Hence the P2 arrangement is used on Full and Plug-in Hybrid Vehicles.

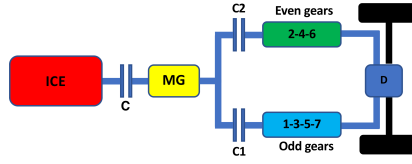


Figure 11: P2 configuration [13]

1.1.10 P2.5

The P2.5 configuration vehicles are equipped with the EMD which is positioned in the gearbox of a dual clutch transmission, attached only to the shaft of even or odd gears.

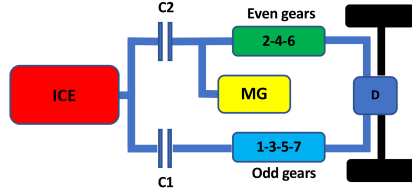


Figure 12: P2.5 configuration [13]

1.1.11 P3

The P3 configuration is similar to the P2, with the difference that the EMD is mounted downstream of the gearbox, directly to the axle. As a consequence, since the gearbox is a speed reducer and a torque amplifier device, generally, the P3 mounted E-motor are bigger than P2 ones because the dimensions of an electric motor mostly depend on torque capability. Thus, the P2 configuration is usually preferred.

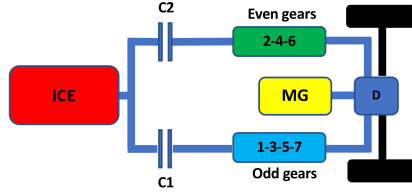


Figure 13: P3 configuration [13]

1.1.12 P4

Also in P4 configuration the EMD is connected to the axle, upstream the differential, but, differently from the P3, to the one independent from the ICE. Thus, these type of vehicles have one axle running on the ICE and the other one, independently, on the EM. Generally the P4 configuration is mounted on Plug-in Hybrid Vehicles.

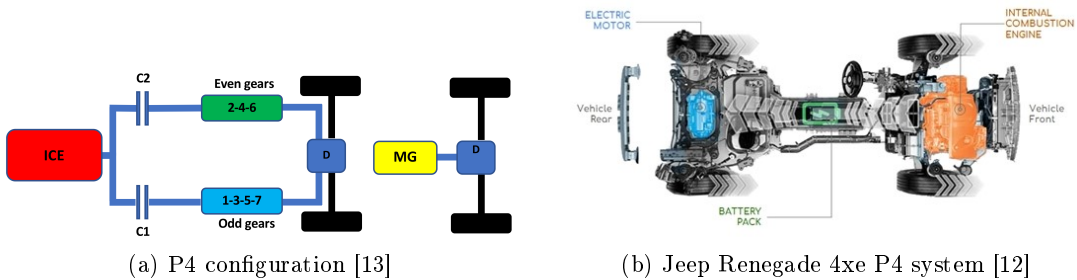


Figure 14

1.1.13 P5

Similar to the P4, P5 configuration is characterized by the presence of the EMD on the independent axle, but connected directly to the wheels, downstream the differential. Two motors must be mounted.

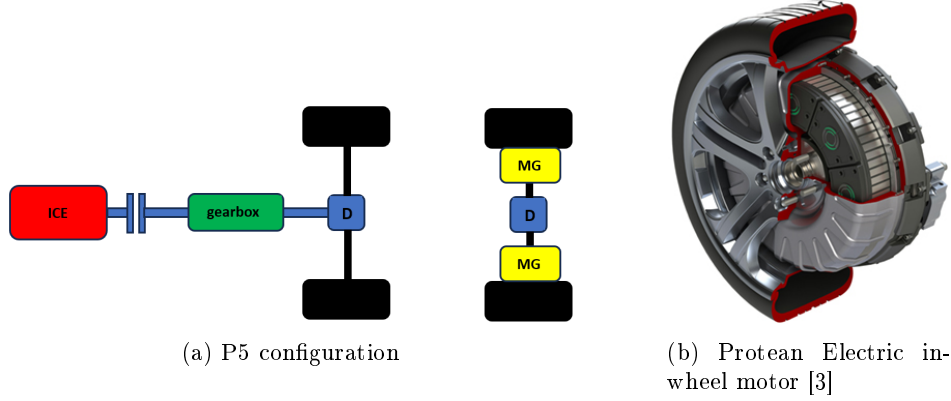


Figure 15

1.1.14 PS

The last category is the PS, one that stands for Power Split. This type of configuration is characterized by 2 EMDs connected by planetary gears and integrated inside the gearbox. This type of solution is mounted on Toyota's Full Hybrid Vehicles.

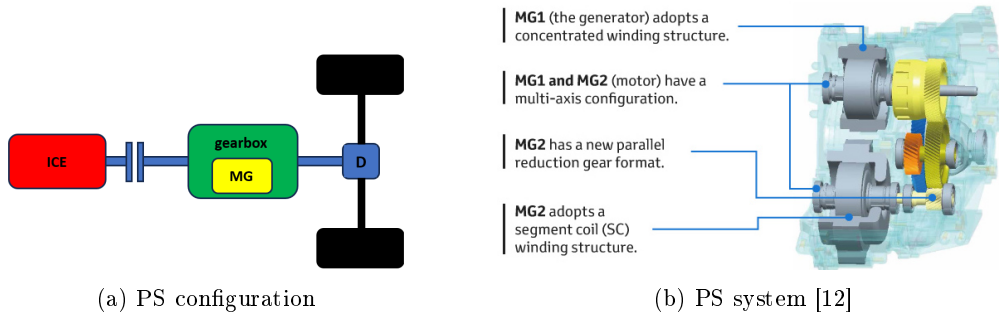


Figure 16

1.2 Electric motor types used in electric powertrains

There is a wide range of choice regarding the type of electric motor when we talk about electric powertrain, and so electric vehicles. E-motors for traction can be divided in two big categories: (i) electric motors with permanent magnets and (ii) magnetless electric motors. The first category includes SPM (Surface Permanent Magnets) motor, IPM (Internal Permanent Magnets) motor and the PMSyRM (PM-assisted Synchronous Reluctance Motor). Instead, the second one includes IM (Induction Motor), SyRM (Synchronous Reluctance Motor) and EESM (Electrically Excited Synchronous Machine).

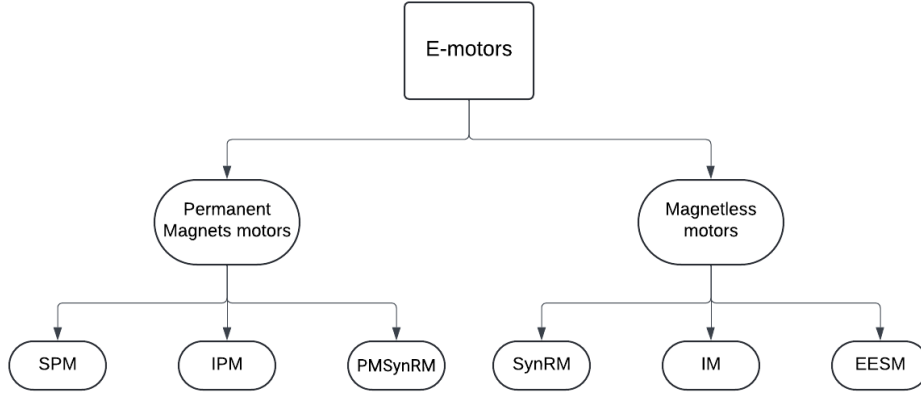


Figure 17: E-motors classification

According to the requirements of the market, the car makers can decide the type of motor to be equipped on the vehicles. They can choose, for example, an induction motor in order to reduce the cost or an IPM motor to obtain a higher efficiency and a higher power density.

There is a characteristic that is common to all these types of motor: they are all 3-phase AC (Alternating Current) motors exploiting a rotating magnetic field to generate a torque.

2 Analysis of electric motor technologies

2.1 Rotating Magnetic Field

In 1888, the Italian electrical engineer Galileo Ferraris explored, with his research published in a paper to the Royal Academy of Sciences in Turin, the possibility of building an AC motor exploiting a rotating magnetic field. The same year, Nikola Tesla obtained a United States patent for a design of a AC electro-magnetic motor based on this principle. From then on, the pioneering work of Galileo Ferraris and Nikola Tesla laid the foundation for the development and widespread adoption of AC electric motors, revolutionizing countless industries and shaping the modern electrical technologies.

In order to understand the different working principles of electric motors, the initial step involves defining what constitutes a rotating magnetic field. Let's start from a simple case, the one depicted in Figure 18.

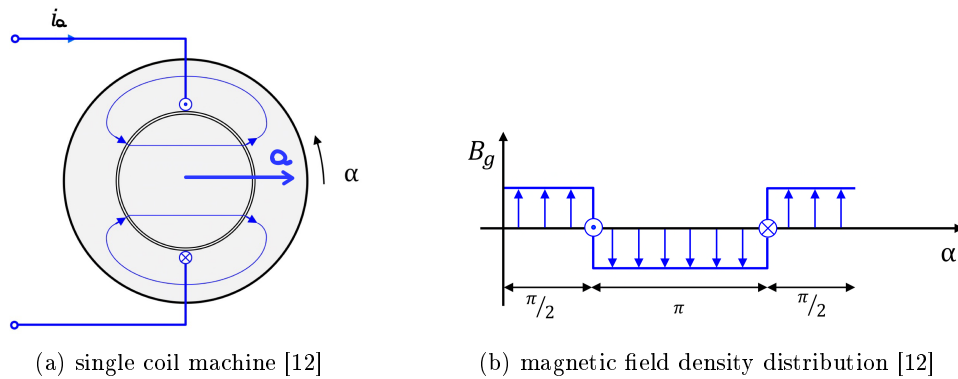


Figure 18

The stator is simply fed by the current of phase a . According to the Biot-Savart law, the current flowing in the windings generates a magnetic field, represented by the blue lines in Figure 18b. By considering a polar coordinate α , we can evaluate the value of the magnetic field as a function of the stator's position. This evaluation leads to the characteristic curve shown in Figure 18b. It can be noticed that, since the windings are concentrated, the magnetic field assumes two values, equal in magnitude and opposite in sign. Moreover the steps, equal to the value of the current multiplied by the number of windings, are in correspondence of the angular position of the windings, i.e. $\alpha = \frac{\pi}{2}$ and $\alpha = \frac{3}{2}\pi$.

Let's now consider a more complex case, in which the windings are distributed.

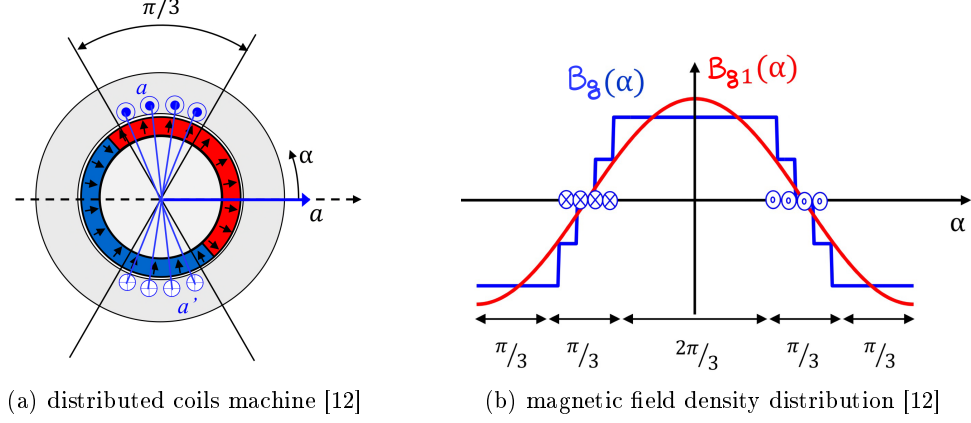


Figure 19

In this case, the stator is still fed by phase a , but the current is distributed into 4 slots, space shifted by $\frac{\pi}{9}$. As before, the magnetic field distribution, according to the angular position α , can be evaluated, remembering that each step is equal to the value of the current multiplied by the number of windings in each slot. The so found characteristic curve has a trend which seems to be more sinusoidal, as shown by the blue graph in Figure 19b. We can even evaluate the first harmonic of the Fourier transform, in red in Figure 19b, in order to better highlight the error between the real distribution and the sinusoidal one. It is noticeable that a higher number of slots means lower steps amplitude and, hence, a magnetic field characteristic closer to the first harmonic.

Since phase a is fed by AC, the current is not constant and the previously represented magnetic field wave is not stationary, but pulsates with a pulsation equal to the electrical one (ω). To grasp this concept refer to Figure 20, which shows the winding current and the magnetic field distribution over the time. Choosing a fixed position on the stator, for instance $\alpha = 0$, we can evaluate the magnetic field, for three different instants of time, in that point. As said before, the current is alternating, hence it assumes three different values for the three instants of time considered.

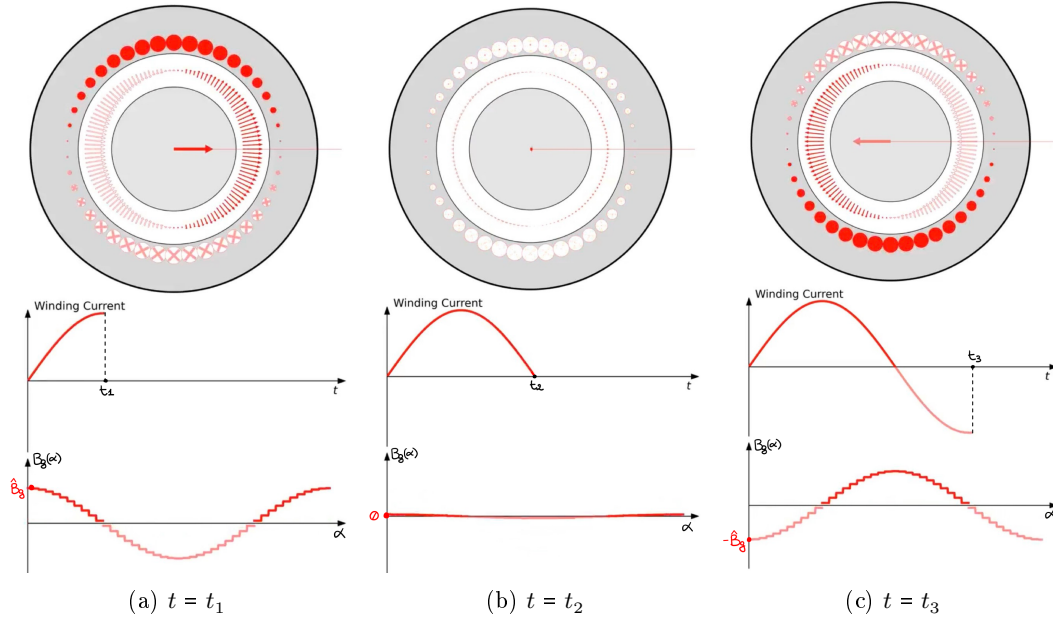


Figure 20

When $t = t_1$, the current assumes the maximum value, as well as B_g , for $\alpha = 0$. The following instant, when $t = t_2$, the current value is zero, hence the magnetic field is null. Instead, when the current assumes the minimum value, at $t = t_3$, the magnetic field is maximum in amplitude and negative in sign. It has been shown that a point in the stator feels a non-constant magnetic field.

The same procedure can be applied for phase b and c .

By superimposing the magnetic fields generated by the three phases, the total magnetic field distribution in the stator can be found. As done before, we evaluate the total magnetic field for three different instants of time, finding the magnetic field wave shown at the bottom of Figure 21.

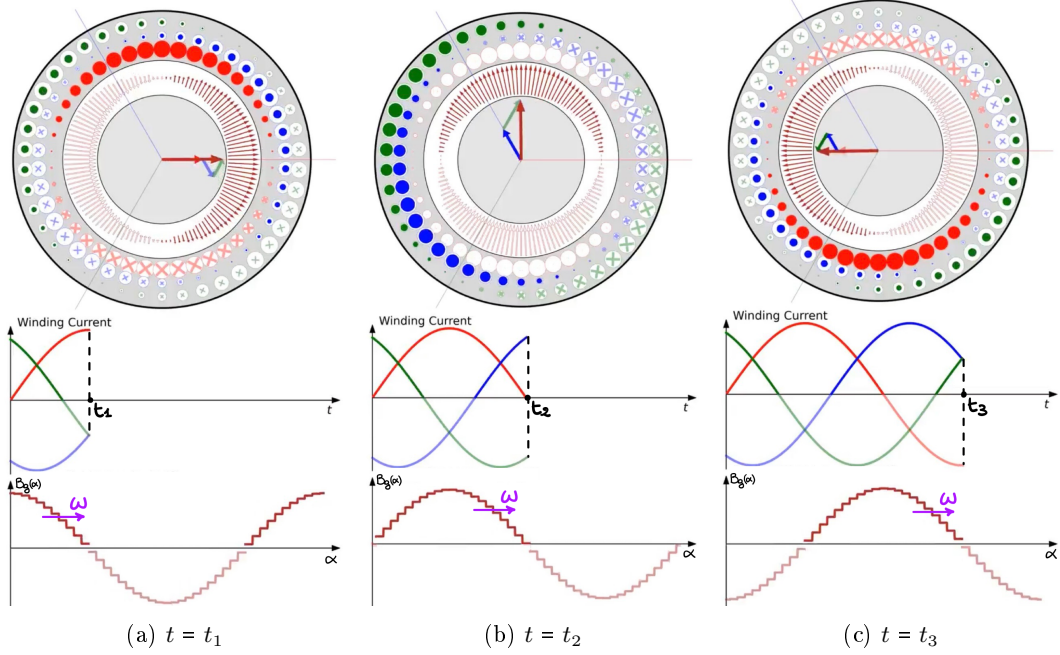


Figure 21

The so obtained result shows that the total magnetic field, generated by the 3-phases currents, does not pulsate, as in the case represented in Figure 20, but it moves with a speed equal to ω . Physically, a moving magnetic wave translates into a rotating magnetic field in the stator.

Once the concept of RMF is understood, we can study how to obtain such field.

2.2 Fundamental concepts of 3-phase system

Firstly, it is of considerable importance to grasp the concept of a 3-phase system and why it is so used, not only in electric motors, but even, for example, in the distribution grid and in the electric power generation based on rotating systems, as hydro-electric and wind turbines.

A 3-phase system is an electrical circuit having three phases i.e. it can be decomposed in three wires. This means that, overall, the circuit can be described by using three voltages and three currents, one for each wire. The three sinusoidal (AC) voltages are equal in amplitude and phase shifted by 120° ; this is true also for the currents.

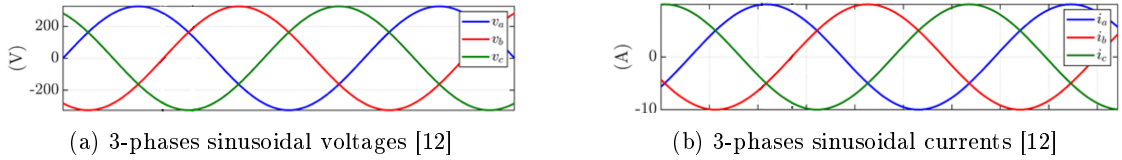


Figure 22

There are many advantages compared to a single-phase system among which: less cabling for equal power transfer, higher compactness and, above all, a constant instantaneous power. This last advantage is really important for an electric motor because, although it works using AC, it can still provide a constant power, which turns out into a constant torque. Moreover, it can be demonstrated that a stator fed by a 3-phase circuit produces a magnetic field that rotates with an angular frequency equal to the electrical pulsation ω , the RMF.

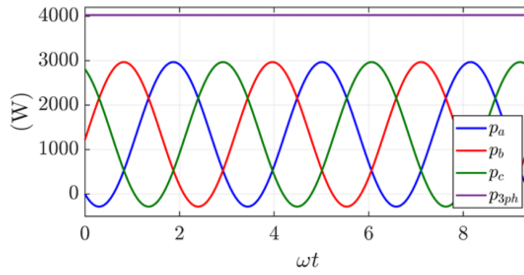


Figure 23: instantaneous and total power of the 3-phases system [12]

After understanding what is the RMF and how it can be generated, we can move on by analyzing, in the following sections, the working principles of the most common types of E-motor used in electric powertrains and how the rotating magnetic field can be exploited in order to produce torque.

2.3 Fundamental concepts of Permanent Magnets Electric Motors

In the automotive industry, permanent magnets electric motors are renowned for their exceptional performance attributed to their high efficiency, compactness and ease of control. However, their notable drawback is their elevated cost, primarily due to the utilization of rare-earth magnets. These magnets can either be surface mounted onto the rotor, in the case of Surface Permanent Magnet (SPM) motors, or integrated within the rotor itself, for Internal Permanent Magnet (IPM) motors. While this distinction may not seem so important, its implications are significant. As we will explore further on, it alters the motor's behavior and performance characteristics.

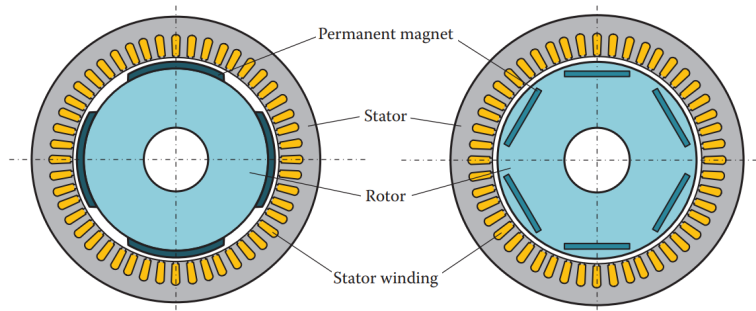


Figure 24: SPM motor (left) and IPM motor (right) [4]

Before going deeper in details, let's first have an overview of the working principle of this two motors. The 3-phase electric motor is composed by two main parts: the stator and the rotor. The stationary stator embeds the windings that produce the RMF (Rotating Magnetic Field). Instead, in the rotor, which is the rotating part, there are the magnets. The important thing to understand, as we discussed before, is that, even if the stator does not move, it produces a magnetic field which is rotating.

The underlying phenomenon that allows the rotor to rotate is very simple. As we have all experienced at least once in our lives, when trying to bring two magnets closer together, we feel an attractive or a repulsion force, depending on the type of poles we are trying to put in contact. In particular, opposite poles will attract each other (North-South), while equal poles will repulse each other (North-North or South-South). By putting the rotor, and so the magnets, inside the stator, they will be affected either by a repulsive or an attractive force, due to the presence of the magnetic field produced by the stator windings, as depicted in Figure 25.

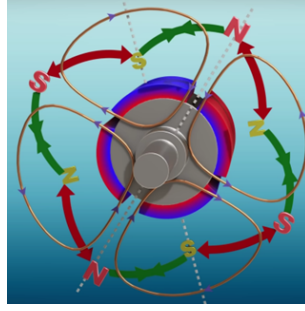


Figure 25: Equilibrium of interaction forces rotor magnets-stator RMF [10]

As a consequence, the rotor will rotate trying to align the poles of the magnets to the opposite ones of the magnetic field. Therefore, since the magnetic field is rotating, even the rotor will rotate, at the same speed, still trying to align the magnets with the field.

2.3.1 Surface-mounted Permanent Magnets (SPM) Motors

In order to understand the differences between SPM and IPM motors, the concept of isotropy has to be introduced. The SPM machine is said to have an isotropic rotor. This implies that the rotor exhibits uniform magnetic properties in all directions. In other words, the magnetic characteristics, such as its reluctance, remain constant regardless of the direction of measurement. The important consequence of isotropy is that the phase inductance L_s is independent from the rotor position. Thus the SPM machine circuit can be represented with 3 equal inductances, as depicted in Figure 26. Later on, in Section 2.3.2, the concept of isotropy and anisotropy will be further investigated.

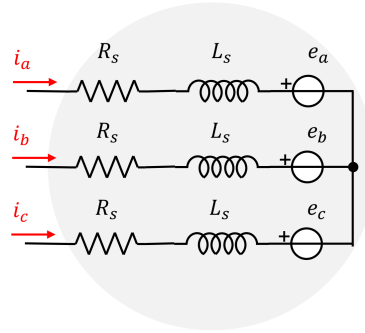


Figure 26: SPM machine circuit [12]

Thanks to the isotropy characteristic of the rotor, the flux linkage vector can be simplified, as shown in the Equation 1.

$$\begin{bmatrix} \lambda_a \\ \lambda_b \\ \lambda_c \end{bmatrix} = \begin{bmatrix} L_a & M_{ab} & M_{ac} \\ M_{ba} & L_b & M_{bc} \\ M_{ca} & M_{cb} & L_c \end{bmatrix} \cdot \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \begin{bmatrix} \lambda_{m,a}(\theta) \\ \lambda_{m,b}(\theta) \\ \lambda_{m,c}(\theta) \end{bmatrix} = L_s \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \lambda_m \begin{bmatrix} \cos(\theta) \\ \cos(\theta - \frac{2\pi}{3}) \\ \cos(\theta + \frac{2\pi}{3}) \end{bmatrix} \quad (1)$$

By using the Clarke transformation, Equation 1 can be simplified, reducing the number of variables from 3 to 2.

$$\begin{bmatrix} \lambda_\alpha \\ \lambda_\beta \end{bmatrix} = L_s \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \lambda_m \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} \quad (2)$$

As last step, in order to further simplify the equation, the rotation transformation can be applied, finding Equation 3.

$$\begin{bmatrix} \lambda_d \\ \lambda_q \end{bmatrix} = L_s \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \lambda_m \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (3)$$

An important thing to remember is that the quantity in the $\alpha\beta$ and dq reference frames are fictitious quantities i.e. they cannot be physically measured. The ones we can measure and that can be controlled are the abc quantities. The reason why $\alpha\beta$ and dq reference frames are used is because they simplify the model and so allow an easier study.

The voltage equation of the SPM motor can be written starting from the general definition of voltage.

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = R_s \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \lambda_a \\ \lambda_b \\ \lambda_c \end{bmatrix} \quad (4)$$

As before, we can simplify Equation 4 by evaluating its equivalent form in the $\alpha\beta$ and dq reference frames, respectively Equation 5 and Equation 6.

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = R_s \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + L_s \frac{d}{dt} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \lambda_m \frac{d\theta}{dt} \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix} \quad (5)$$

$$\begin{bmatrix} v_d \\ v_q \end{bmatrix} = R_s \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \lambda_d \\ \lambda_q \end{bmatrix} + \omega \begin{bmatrix} -\lambda_q \\ \lambda_d \end{bmatrix} \quad (6)$$

Starting from the general equation of torque for a 3-phase electric motor (Equation 7), we can deduce the one typical of SPM motors.

$$T_e = \frac{3}{2}p(\lambda_d i_q - \lambda_q i_d) \quad (7)$$

By substituting the values of λ_d and λ_q with the ones found in Equation 3, the torque equation of a SPM machine is found:

$$T_e = \frac{3}{2}p(\cancel{L_s i_d i_q} + \lambda_m i_q - \cancel{L_s i_d i_q}) \rightarrow T_e = \frac{3}{2}p\lambda_m i_q \quad (8)$$

p is the number of pole pairs and λ_m is the PM (Permanent Magnet) flux linkage.

By analyzing Equation 8, it can be noticed that i_d does not contribute to torque generation. Moreover, the electromagnetic torque is proportional to i_q . From this considerations we can evaluate the MTPA (Maximum Torque Per Ampere) of the SPM motor i.e. the value of γ , that is the angle of the space vector $\mathbf{i}_{dq}$ in the dq reference frame, which maximizes the torque, given the current amplitude.

In order to find γ , we firstly have to transform the electric torque function into an equivalent one expressing the variable γ . Since, as said before, it is the characteristic angle of the vector $\mathbf{i}_{dq}$, i_q can be expressed as follow:

$$i_q = |\mathbf{i}_{dq}| \sin \gamma \quad (9)$$

So, by substituting Equation 9 into Equation 8, the electromagnetic torque as function of the angle γ is obtained.

$$T_e = \frac{3}{2}p\lambda_m |\mathbf{i}_{dq}| \sin \gamma \quad (10)$$

We can easily evaluate the value of γ which maximizes the torque, which is $\frac{\pi}{2}$. In other words, in the dq plane, the current vector $\mathbf{i}_{dq}$ is aligned with the q axis, i.e. the MTPA for a SPM motor is the q axis.

2.3.2 Internal-mounted Permanent Magnets (IPM) Motors

Differently from the SPM, the IPM motors have an anisotropic rotor, which leads to different magnetization properties depending on the direction of measurement. The effect of the anisotropic rotor on the magnetic field is shown in Figure 27a and 27b.

In Figure 27 the rotor with the magnets (represented in green and red) and the magnetic field produced by the windings, in this case the one of phase a , is depicted. In Figure 27a, when the q axis is aligned to phase a , the magnetic field induced by the windings only passes through the stator iron, characterized by a relatively low value of reluctance. Instead, in Figure 27b, when the d axis is aligned to phase a , the flux passes both through the stator iron and the magnets. A more precise analysis is represented in Figure 28, where, through a FEM evaluation, the field lines are evaluated for the two different configurations.

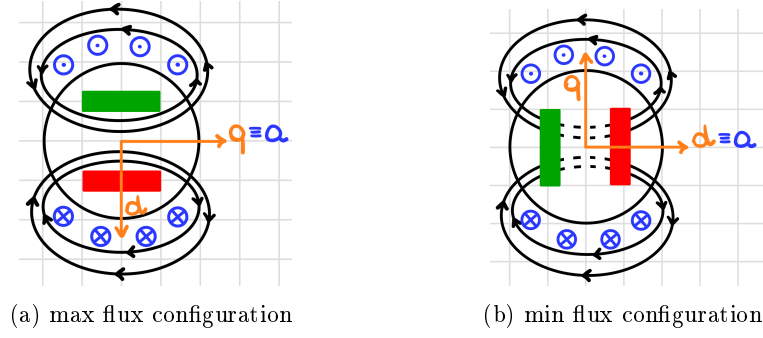


Figure 27

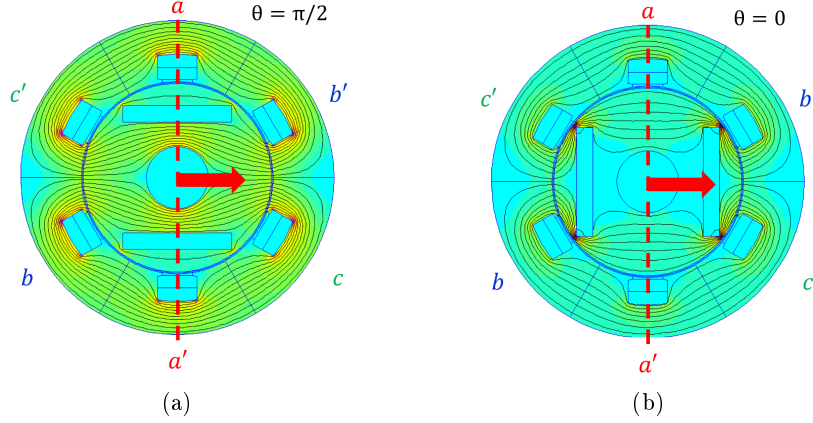


Figure 28: FEM flux analysis when $\theta = 0, \frac{\pi}{2}$ [12]

Since the reluctance of the magnets is pretty high compared to the one of the rotor iron, the resultant magnetic field, and consequently the inductance, will be lower in the condition depicted in Figure 27b. Thus the minimum inductance corresponds to the d axis and it is called L_d , while the maximum inductance, called L_q , corresponds to the q axis. Hence, since the rotor rotates, the value of the inductance of a phase oscillates between L_d and L_q .

The inductance of phase a as function of rotor position θ is shown in Figure 29.

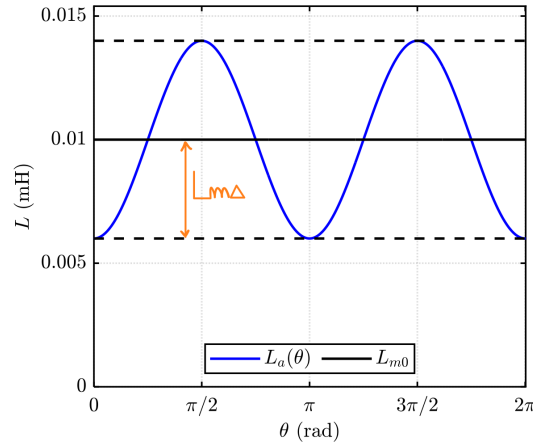


Figure 29: Inductance function [12]

As shown in Figure 29, the inductance varies sinusoidally according to Equation 11, where L_{m0} is the average inductance and $L_{m\Delta}$ is the differential term of anisotropy.

$$L_{ma} = L_{m0} + L_{m\Delta} \cos(2\theta) \quad (11)$$

The self-inductance waveforms of phases b and c are phase shifted by 120° with respect to L_{ma} . Hence Equations 12 and 13 are deduced.

$$L_{mb} = L_{m0} + L_{m\Delta} \cos(2\theta + \frac{2}{3}\pi) \quad (12)$$

$$L_{mc} = L_{m0} + L_{m\Delta} \cos(2\theta - \frac{2}{3}\pi) \quad (13)$$

As we will explore further later, this peculiarity will add the reluctance torque in the electromagnetic torque equation.

As done for the SPM machine, the flux linkage vector for an IPM motor can be written.

$$\begin{bmatrix} \lambda_a \\ \lambda_b \\ \lambda_c \end{bmatrix} = \begin{bmatrix} L_a(2\theta) & M_{ab}(2\theta) & M_{ac}(2\theta) \\ M_{ba}(2\theta) & L_b(2\theta) & M_{bc}(2\theta) \\ M_{ca}(2\theta) & M_{cb}(2\theta) & L_c(2\theta) \end{bmatrix} \cdot \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \lambda_m \begin{bmatrix} \cos(\theta) \\ \cos(\theta - \frac{2\pi}{3}) \\ \cos(\theta + \frac{2\pi}{3}) \end{bmatrix} \quad (14)$$

As we discussed before, the inductance matrix does not have constant values as the SPM one, but values which are function of 2θ . Consequently, the magnetic circuit has non constant inductance values.

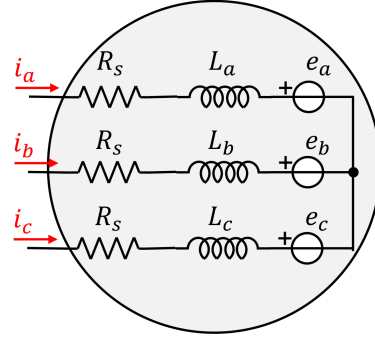


Figure 30: IPM machine circuit [12]

By applying the Park transformation (Clarke + Rotation), we find the equation of flux linkage in the dq reference frame.

$$\begin{bmatrix} \lambda_d \\ \lambda_q \end{bmatrix} = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} + \lambda_m \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (15)$$

Regarding the voltage equation, it is the same as for the isotropic SPM machine, i.e. Equation 6.

We can even evaluate the torque following the same procedure used for SPM: we replace Equation 14 into Equation 7 in order to find the equation of the electromagnetic torque for an IPM motor.

$$\begin{aligned} T_e &= \frac{3}{2}p(\lambda_d i_q - \lambda_q i_d) = \frac{3}{2}p((L_d i_d + \lambda_m) i_q - L_q i_q i_d) = \frac{3}{2}p(L_d i_d i_q + \lambda_m i_q - L_q i_q i_d) = \\ &= \frac{3}{2}p(\lambda_m i_q + (L_d - L_q) i_d i_q) \rightarrow T_e = \underbrace{\frac{3}{2}p\lambda_m i_q}_{\text{PM torque}} + \underbrace{\frac{3}{2}p(L_d - L_q) i_d i_q}_{\text{reluctance torque}} \end{aligned} \quad (16)$$

It is notable that, in addition to the PM torque, which was already accounted for in the SPM torque equation, in the IPM one there is even the contribution of the reluctance torque. Thus, to achieve a higher torque, the difference between L_d and L_q has to be maximized. Practically, this leads to design a rotor with the highest possible anisotropy. Achieving this involves manipulating the rotor's geometry and introducing air cavities, as air exhibits a notably high reluctance value.

By using Equation 9 and Equation 17 we can express the electromagnetic torque as function of γ in order to find the MTPA locus.

$$i_d = |i_{dq}| \cos \gamma \quad (17)$$

$$T_e = \frac{3}{2}p(\lambda_m |i_{dq}| \sin \gamma + (L_d - L_q) |i_{dq}|^2 \sin \gamma \cos \gamma) \quad (18)$$

The MTPA locus of the IPM machine is complicated compared to the SPM case because both i_d and i_q are in play, hence it is not linear. By solving Equation 18 with numerical methods, the MTPA current angle is found. It is in the second quadrant and varies with the current amplitude and the torque. This non-linearity leads to a more complex control. In Figure 31 the MTPA locus of the Tesla Model 3 motor, obtained through a FEA, is depicted in red.

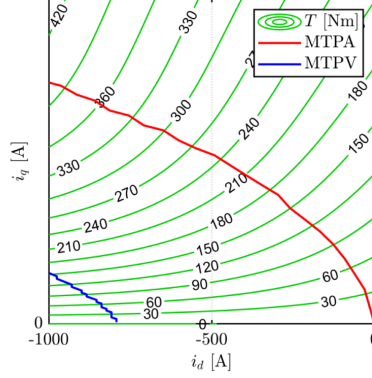


Figure 31: MTPA locus of the Tesla Model 3 motor [12]

Through the analysis of the MTPA curve, the values of i_d and i_q which maximize the torque, for a certain value of the current amplitude, can be determined, as shown in the example depicted in Figure 32. For a torque value of 210 Nm, the corresponding i_d and i_q values are evaluated by projecting the operating point onto the respective axis. Moreover, the two values can be even found by using the chart in Figure 32b.

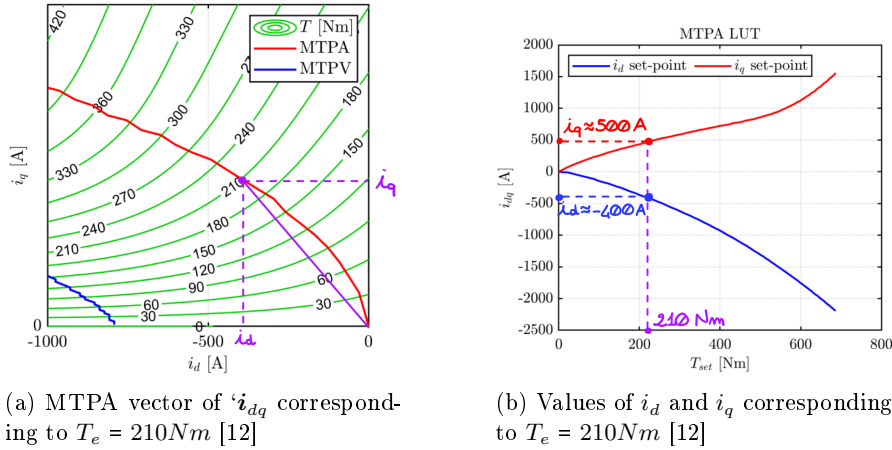


Figure 32

2.4 Electric motors without magnets

2.4.1 Synchronous Reluctance Motors

By introducing Equation 16, we stated that the torque of an electric motor is given by the sum of two components: the PM torque and the reluctance torque. Consequently, a motor with $\lambda_m = 0$, i.e. without magnets, might ideally be designed. The motor is still able to provide a torque, which depends exclusively on the reluctance component. This is the principle on which the Synchronous Reluctance Motor (SynRM) is based. In light of these considerations, the flux linkage and torque equations of a SynRM are the following ones:

$$\begin{bmatrix} \lambda_d \\ \lambda_q \end{bmatrix} = \begin{bmatrix} L_d & 0 \\ 0 & L_q \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \quad (19)$$

$$T_e = \frac{3}{2}p(\lambda_d i_q - \lambda_q i_d) \rightarrow T_e = \frac{3}{2}p(L_d - L_q)i_d i_q \quad (20)$$

As we have already said before speaking about IPM motors, in order to achieve a higher reluctance torque, the design of the rotor should be done in such a way to obtain the highest

possible difference between inductances along d and q axis. In Figure 33 an example of rotor geometry suitable for this type of motor is shown.



Figure 33: Synchronous Reluctance Motor's rotor

As done for SPM and IPM machines, by substituting Equation 9 and Equation 17 into Equation 20, we can evaluate the MTPA for the SynRM.

$$T_e = \frac{3}{2}p(L_d - L_q)|i_{dq}|^2 \sin \gamma \cos \gamma = \frac{3}{2}p(L_d - L_q)|i_{dq}|^2 \frac{\sin(2\gamma)}{2} \quad (21)$$

The maximum value of T_e is obtained when $\gamma = \frac{\pi}{4}$. Hence the current vector $|i_{dq}|$ is positioned in the first quadrant.

2.4.2 Electrically Excited Synchronous Machine

As we have discussed before, a magnetic field in the rotor leads to the interaction with the one produced by the stator. This interaction produces torque. In the previous sections we focused on the analysis of motors where the magnetic field in the rotor, if present, was produced by magnets.

Instead, in this section, a motor where the rotor magnetic field is produced by windings flowed by a current is analyzed.

This type of machine is called Electrically Excited Synchronous Machine (EESM), also known as Wound Field Synchronous Motor (WFSM).

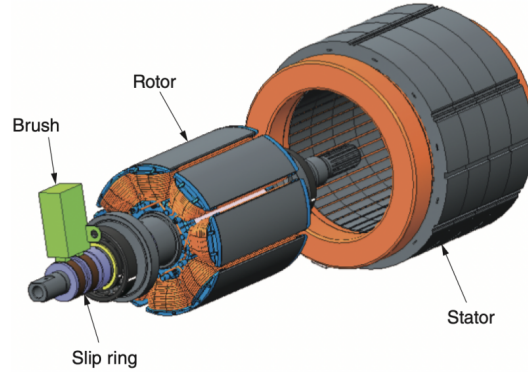


Figure 34: EESM architecture [11]

The rotor windings are supplied in DC. The main advantage of having the windings instead of the magnets is that the rotor excitation field can be modified simply by adjusting the rotor current. Obviously, the PMSMs do not have this possibility since the value of the magnet flux density (λ_m) is almost constant.

On the other hand, there are some drawbacks. The first one, probably the most limiting, is the need to bring current to rotating parts. Usually this task is accomplished by stationary carbon brushes in contact with rotating slip rings. This technology generally arises problems concerning reliability and wear. The second important drawback is the limited cooling capability. The presence of a current in the rotor produces additional losses due to the Joule effect. The design of a cooling system providing the heat extraction from the rotor is more complex and may become a limiting factor.

The EESM has two possible rotor configurations: salient pole and non-salient pole (also called "cylindrical rotor"). The two configurations are shown in Figure 35.

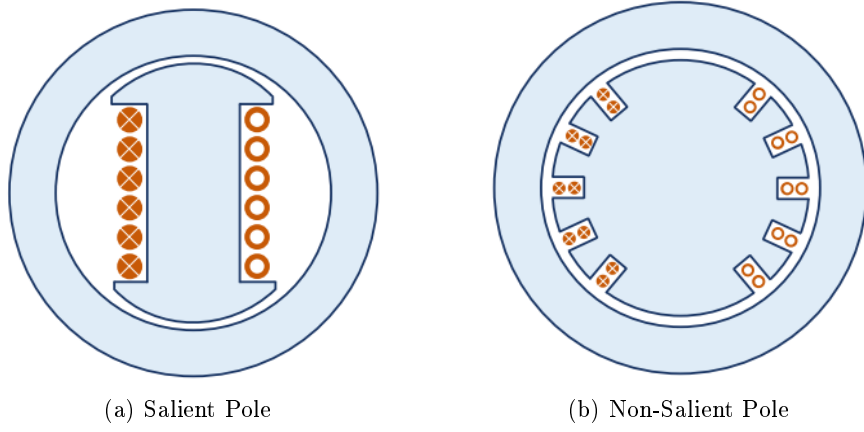


Figure 35: EESM topologies [9]

The salient pole configuration is characterized by a variable airgap, which translates into an anisotropic rotor. As we have already discussed, the rotor anisotropy leads to the generation of a reluctance torque. On the other hand, the cylindrical rotor configuration presents a symmetric rotor geometry and an almost constant airgap, hence the rotor is isotropy. As a consequence, reluctance torque does not contribute to the torque equation.

Since in electrical powertrains for automotive application the salient-pole topology is more widespread, we will focus our attention on this configuration.

The magnetic model of the EESM presents an additional degree of freedom, with respect to the one of others motors, due to the fact that the rotor current can be controlled.

$$\mathbf{v} = \begin{bmatrix} v_d \\ v_q \\ v_f \end{bmatrix}, \mathbf{i} = \begin{bmatrix} i_d \\ i_q \\ i_f \end{bmatrix}, \boldsymbol{\lambda} = \begin{bmatrix} \lambda_d \\ \lambda_q \\ \lambda_f \end{bmatrix}$$

The dq model, in steady state conditions, can be written in the following compact form:

$$\begin{aligned} \mathbf{v} &= \mathbf{R}\mathbf{i} + \boldsymbol{\Omega}\boldsymbol{\lambda} \\ \boldsymbol{\lambda} &= \mathbf{L}\mathbf{i} + \mathbf{M}\mathbf{i} \end{aligned} \quad (22)$$

The resistance, speed, inductance and mutual inductance matrices are reported in Equation 23 and 24

$$\mathbf{R} = \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_f \end{bmatrix}, \quad \boldsymbol{\Omega} = \begin{bmatrix} 0 & -\omega & 0 \\ \omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (23)$$

$$\mathbf{L} = \begin{bmatrix} L_d & 0 & 0 \\ 0 & L_q & 0 \\ 0 & 0 & L_f \end{bmatrix}, \quad \mathbf{M} = \begin{bmatrix} 0 & 0 & M_{sr} \\ 0 & 0 & 0 \\ M_{sr} & 0 & 0 \end{bmatrix} \quad (24)$$

where R_f and L_f are respectively the rotor resistance and inductance, while M_{sr} is the mutual inductance between stator and rotor.

The system of equations corresponding to the dq model introduced in Equation 22 can be written as

$$\begin{cases} v_d = R_s \cdot i_d - \omega \cdot \lambda_q \\ v_q = R_s \cdot i_q + \omega \cdot \lambda_d \\ v_f = R_f \cdot i_f \end{cases} \quad \begin{cases} \lambda_d = L_d \cdot i_d + M_{sr} \cdot i_f \\ \lambda_q = L_q \cdot i_q \\ \lambda_f = L_f \cdot i_f + M_{sr} \cdot i_d \end{cases} \quad (25)$$

Starting from the general equation of electromagnetic torque, which is valid for all Synchronous Machines, the peculiar one of Electrically Excited Synchronous Machine can be deduced. By replacing Equation 25 into Equation 7 the following equation is obtained:

$$\begin{aligned} T_e &= \frac{3}{2}p(\lambda_d i_q - \lambda_q i_d) = \frac{3}{2}p((L_d i_d + M_{sr} i_f) i_q - L_q i_q i_d) = \frac{3}{2}p(L_d i_d i_q + M_{sr} i_f i_q - L_q i_q i_d) = \\ &= \frac{3}{2}p(M_{sr} i_f i_q + (L_d - L_q) i_d i_q) \rightarrow T_e = \underbrace{\frac{3}{2}p M_{sr} i_f i_q}_{\text{excitation field torque}} + \underbrace{\frac{3}{2}p (L_d - L_q) i_d i_q}_{\text{reluctance torque}} \end{aligned} \quad (26)$$

The evaluated torque equation is composed of two terms: the excitation field torque and the reluctance torque. Differently from IPM, where the PM flux cannot be changed, in the EESM the amplitude of the excitation field torque can be controlled by modifying the value of rotor current i_f . As a consequence, different operating conditions can occur: (i) supposing zero rotor current, torque production is handled only by reluctance torque, and so by rotor anisotropy, hence the machine behaves like a synchronous reluctance motor (SyRM). (ii) increasing rotor current leads to the appearance of excitation field torque; the machine behaves as an IPM motor. (iii) having $L_d \approx L_q$, for instance for cylindrical rotor configuration or if machine saturation occurs due to high rotor current, the reluctance torque is null, hence the machine behaves like a SPM motor.

3 Drives with Electrically Excited Synchronous Machine

The introduction of a third degree of freedom makes the control of EESM more complicated with respect to one of PMSM, where the variables that have to be controlled are only two. In response to this complexity, several control strategies can be implemented to manage and optimize the performance of EESM in different operating conditions.

- Stator MTPA: used to achieve the maximum torque for a given stator current
- Rotor MTPA: used to achieve a certain torque with the minimum value of rotor current. Consequently, if the required torque can be provided only by the reluctance component, the rotor current is set to zero. The stator currents are chosen according to the Stator MTPA.
- Loss Minimizing: aims to minimize losses, both in stator and rotor, hence maximizing the efficiency.
- Zero d -Axis Current: the current vector is set in the q direction, hence $i_d = 0$. This strategy simplifies the control, reducing the degrees of freedom from three to two. The remaining variables are set by minimizing the losses.
- Unity Power Factor (UPF): maximize the power factor to fully exploit the power capability of the inverter.

3.1 PLECS simulation of a torque/current control scheme

A PLECS model of an Electrically Excited Synchronous Machine has been used to implement a simple torque/current control scheme based on a controlled voltage generator. The aforementioned scheme is shown in Figure 36. The control strategy used is the Stator MTPA.

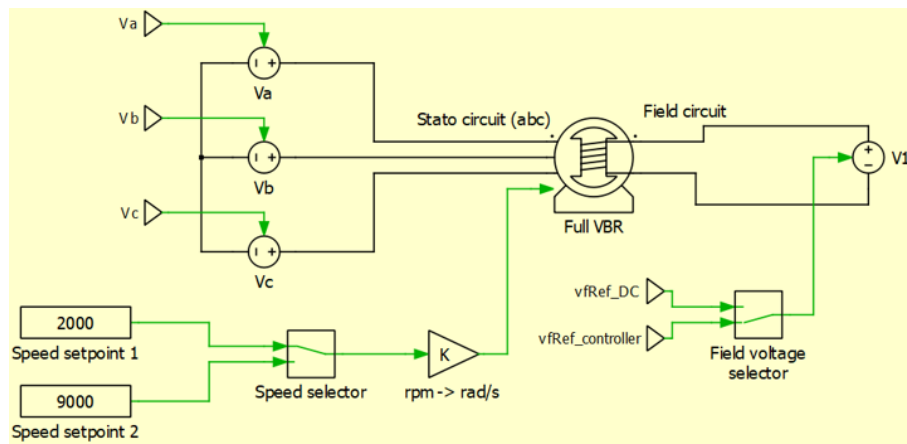


Figure 36: torque/current control scheme

Depending on the rotational speed of the motor (ω_r), two constant reference values of excitation current are used. In particular, $i_{f,ref} = 5$ A for high speed, while $i_{f,ref} = 15$ A for low speed. The rotational speed is set to $n = 2000$ rpm or $n = 9000$ rpm through the dedicated blocks represented in Figure 36.

The block implementing the values of the excitation current is shown in Figure 37.

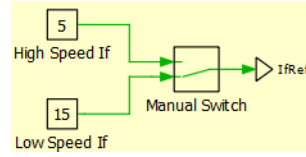


Figure 37: Excitation current block

Furthermore, the control can be performed by choosing between two values of reference field excitation voltage: one represents the open loop voltage, while the other is the voltage obtained through the control block depicted in Figure 38.

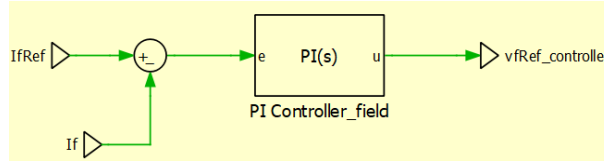


Figure 38: PI controller blocks for the evaluation of $V_{f,ref}$

Once the values of rotational speed and current are set, a reference torque can be established.

After setting the reference current and torque, a dedicated block, depicted in Figure 39, automatically determines the reference currents $i_{d,ref}$ and $i_{q,ref}$ that should be provided to the motor in order to achieve the desired torque.

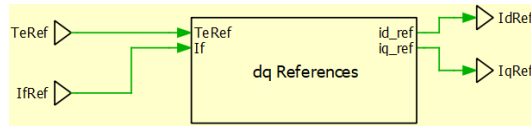


Figure 39: $i_{d,ref}$ and $i_{q,ref}$ block

The control of the motor is achieved by imposing to the controlled generator the values of voltages (V_a , V_b , V_c) that allow the motor to produce the desired torque. The evaluation of these voltages is done through a PI controller having the difference between the reference and the measured current as input and the reference voltage as output. For sake of simplicity the evaluation of the reference voltages is done in the dq reference frame.

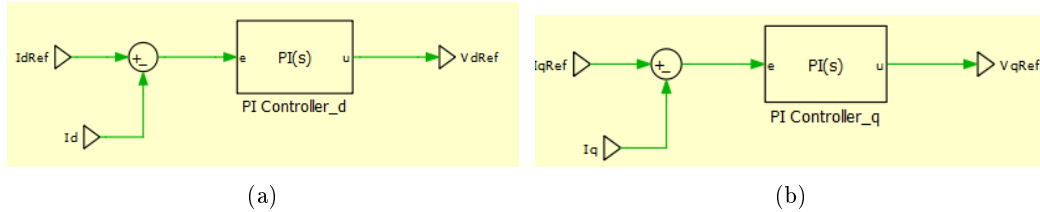


Figure 40: PI controller blocks for the evaluation of $V_{d,ref}$ and $V_{q,ref}$

The values of i_d and i_q can be determined starting from the measurement of the three phase currents i_a , i_b and i_c . Through a Park transformation block, the currents in the dq reference frame are found.

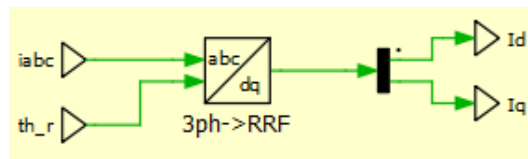


Figure 41: abc to dq current transformation block

Once the reference voltages in the dq reference frame are obtained, through a block implementing the reverse Park transformation, the controlling voltages V_a , V_b , V_c can be evaluated.

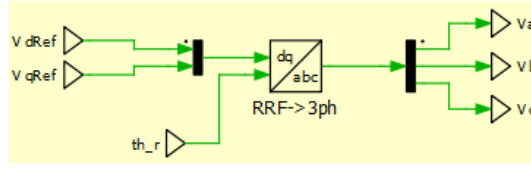


Figure 42: dq to abc voltage transformation block

A correct tuning of the PI controller is done with an iterative method. Initially, tentative values of K_i and K_p are arbitrary chosen. To determine if selected values are correct, the convergence of measured values to reference ones is evaluated. If the functions converge, the PI controller is properly tuned, and further adjustments to K_i and K_p are not needed. Otherwise, if convergence is not satisfied, K_i and K_p must be modified until proper values that ensure convergence are found.

3.2 Control results

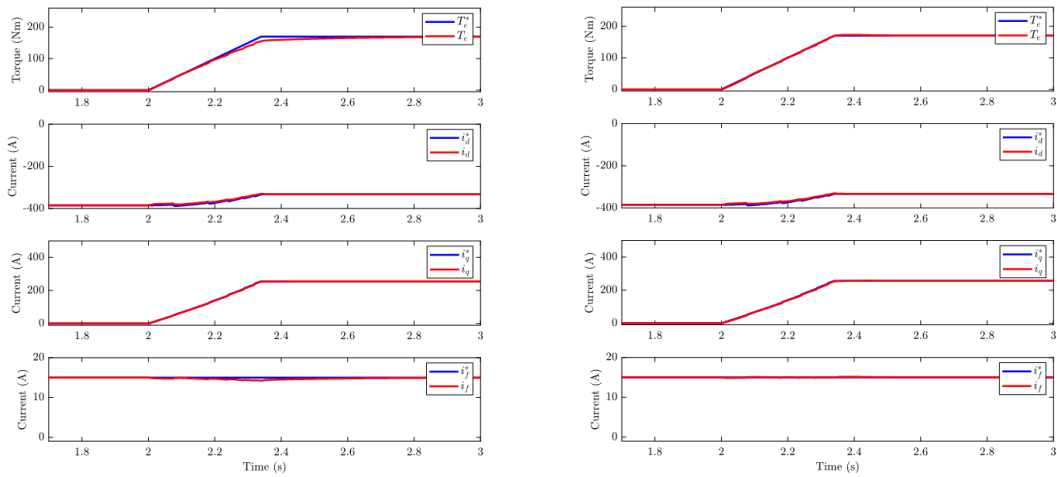
The results obtained from the control scheme explained in Section 3.1 are reported in this final section. In particular, two types of analysis are conducted, one at low speed and the other one at high speed. For each condition, two types of control are performed: one using the open-loop reference voltage, and the other using the reference voltage obtained from the controller. The reference torque is set to two values: $T_{e,ref} = 170$ Nm and $T_{e,ref} = 250$ Nm.

As previously mentioned, the excitation current values are set to $i_{f,ref} = 5$ A and $i_{f,ref} = 15$ A, depending on the rotational speed.

3.2.1 Low speed control

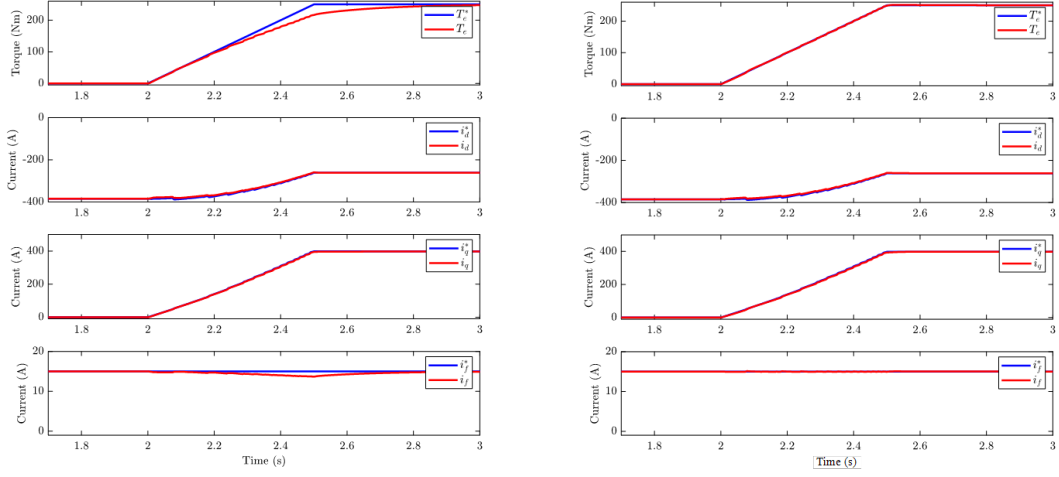
The torque and current graphs obtained, at low speed, by controlling the motor at constant excitation current and in MTPA conditions, for the two values of reference torque and reference excitation voltage, are represented in Figure 43 and Figure 44.

The blue lines represent the reference values, so the ones that we want to obtain, while the red lines represent the measured values.



(a) Open-loop reference excitation voltage control (b) Controlled reference excitation voltage control

Figure 43: Low speed, $T_{e,ref} = 170$ Nm



(a) Open-loop reference excitation voltage control (b) Controlled reference excitation voltage control

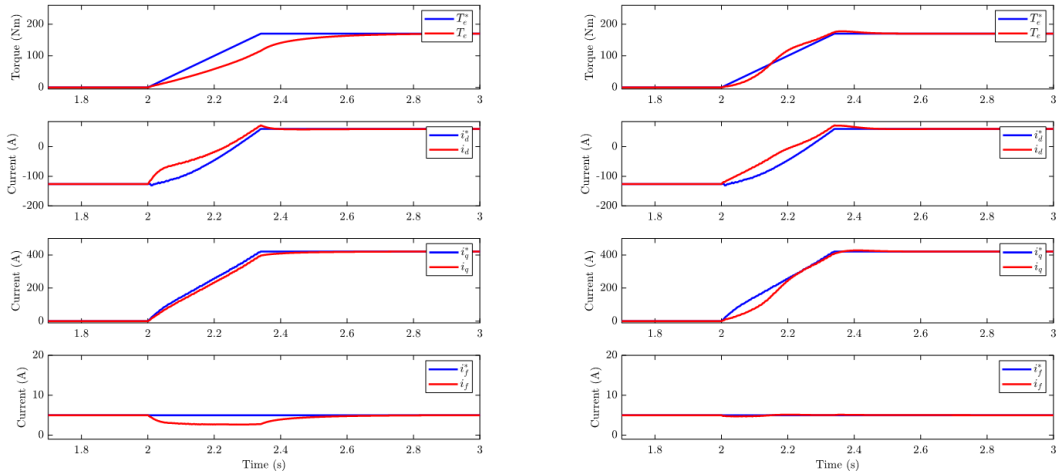
Figure 44: Low speed, $T_{e,ref} = 250$ Nm

The graphs show that, as expected, by increasing the torque output, the values of i_d and i_q increase too. Knowing the values of these currents, and so the components of the current vector in the dq reference frame, its magnitude can be evaluated. We find that when $T_{e,ref} = 170$ Nm the current vector $|\mathbf{i}_{dq}| = 419.42$ A, while for $T_{e,ref} = 250$ Nm, $|\mathbf{i}_{dq}| = 476.12$ A.

Furthermore, it is notable that a better control is achieved when the reference excitation voltage is set equal to the controlled one. This approach results in a reduced error between the measured and reference values.

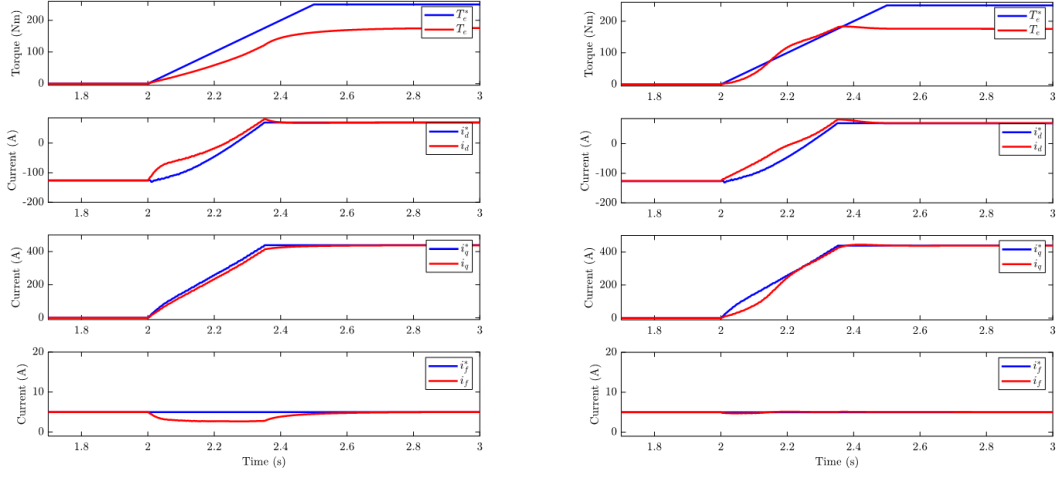
3.2.2 High speed control

In Figure 45 and Figure 46 the graphs of current and torque obtained at high speed and with a low value of excitation current are reported.



(a) Open-loop reference excitation voltage control (b) Controlled reference excitation voltage control

Figure 45: High speed, $T_{e,ref} = 170$ Nm



(a) Open-loop reference excitation voltage control (b) Controlled reference excitation voltage control

Figure 46: High speed, $T_{e,ref} = 250$ Nm

As done for the low speed control, the current vector magnitude can be evaluated. When $T_{e,ref} = 170$ Nm the current vector $|\mathbf{i}_{dq}| = 426.98$ A, while for $T_{e,ref} = 250$ Nm, $|\mathbf{i}_{dq}| = 443.37$ A.

It is important to notice that at high speed the reference torque of 250 Nm cannot be achieved. This can be seen in Figure 46, where the measured torque at steady state is lower than the reference torque. This discrepancy occurs because at $n = 9000$ rpm the motor working point is located in the flux weakening region, resulting in a maximum torque output lower than the nominal value. As a consequence, i_d and i_q are saturated at the maximum values permitted at this rotational speed.

Therefore, at 9000 rpm, with an excitation current equal to 5A, the maximum torque the motor can produce is $T_{e,max} = 175.66$ Nm.

References

- [1] URL: https://www.researchgate.net/publication/321093822_Intelligent_Energy_Management_Control_for_Extended_Range_Electric_Vehicles_Based_on_Dynamic_Programming_and_Neural_Network.
- [2] URL: https://www.researchgate.net/figure/Topology-of-hybrid-drive-systems-based-on-23-P0-electric-motor-is-used-only-to-start_fig1_367071291.
- [3] URL: <https://chargedevs.com/newswire/two-new-in-wheel-electric-drive-systems-unveiled/>.
- [4] Ali Emadi. *Advanced Electric Drive Vehicles*. CRC Press Taylor & Francis Group, 2015.
- [5] US department of energy. *How Do All-Electric Cars Work?* URL: <https://afdc.energy.gov/vehicles/how-do-all-electric-cars-work>.
- [6] US department of energy. *How Do Fuel Cell Electric Vehicles Work Using Hydrogen?* URL: <https://afdc.energy.gov/vehicles/how-do-fuel-cell-electric-cars-work>.
- [7] US department of energy. *How Do Hybrid Electric Cars Work?* URL: <https://afdc.energy.gov/vehicles/how-do-hybrid-electric-cars-work>.
- [8] US department of energy. *How Do Plug-In Hybrid Electric Cars Work?* URL: <https://afdc.energy.gov/vehicles/how-do-plug-in-hybrid-electric-cars-work>.
- [9] Federica Graffeo. *Modeling and Design of Wound-Field Synchronous Machines for Transportation Applications*. Doctoral Program in Electrical Engineering, 2023.
- [10] Lesics. *Tesla Model 3's motor - The Brilliant Engineering behind it*. URL: https://www.youtube.com/watch?v=esUb7Zy50io&t=196s&ab_channel=Lesics.
- [11] Nicola Macrì. *High-Performance Torque Control of Electrical Excited Synchronous Motors*. Master Degree in Mechatronic Engineering, 2024.
- [12] Gianmario Pellegrino. *Electrical Machines*. Bachelor Degree in Automotive Engineering, Politecnico di Torino, 2023.
- [13] Andrea Tonoli. *Motor Vehicle Design*. Bachelor Degree in Automotive Engineering, Politecnico di Torino, 2023.